

Shape Optimization

Lab sheets: Sessions 1–11

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English teaching draft

Each lab lasts 90 minutes. Times include discussion and feedback. Optional extensions are homework or replacements for an early finish; they are not additional required work within the 90 minutes. Each lab also has two supplementary programming exercises in a Python Jupyter notebook, with statements, starter code and checks. These can be assigned as homework or a separate computer session; see [the notebook index](#) and [setup instructions](#).

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Bring a ruler and, for Lab 2, a compass. A calculator is optional. Write down the feasible set before computing. When proposing an optimum, explain both why it is admissible and why no competitor gives a better value.

1 Models and elementary distance problems

Lab 1 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 1.1 (Formulate before solving — 15 minutes). For each problem, specify the variable, admissible class and objective. State whether the request is a minimum or a maximum.

- (a) Choose a point of a line closest to a fixed point Q off the line.
- (b) Use a loop of string of length 12 to enclose as much area as possible.
- (c) Join A to B by a curve that touches a given line ℓ .

For (b), compare the areas enclosed by an equilateral triangle, a square and a circle. Explain why these three computations do not prove optimality among all closed curves.

Exercise 1.2 (Projection — 20 minutes). Let H be the orthogonal projection of Q onto a line ℓ . Show that $|Q - X| \geq |Q - H|$ for every $X \in \ell$, and identify the equality case. Apply this to $Q = (1, 3)$ and the horizontal axis.

Now replace the whole horizontal axis by the segment from $(2, 0)$ to $(5, 0)$. Where is the closest point? Explain why the foot of the perpendicular is no longer the answer.

Exercise 1.3 (Heron's reflection problem — 30 minutes). Let $A = (0, 2)$ and $B = (6, 4)$; a path from A to B must touch the horizontal axis. Reflect B across that axis to B' .

- (a) Explain why a shortest candidate can be sought among broken segments $A-X-B$ with X on the axis.
- (b) Show $|A - X| + |X - B| = |A - X| + |X - B'| \geq |A - B'|$.
- (c) Find where $[A, B']$ crosses the axis and compute the minimum length.
- (d) Draw the equal-angle reflection property at the optimal contact point.

Exercise 1.4 (How constraints change a model — 15 minutes). (a) What is the infimum of the perimeter of disks with positive radius? Is it attained within this class?

- (b) What changes when their area is fixed at 9π ?
- (c) If a numerical search reports a best radius from a finite list, which comparison has actually been made?

Synthesis — 10 minutes. For the reflection problem, identify the bound valid for every competitor and the competitor attaining it. Explain the roles of feasibility and equality in the proof.

Optional extension. Solve Heron's problem for arbitrary points $A = (a, h)$ and $B = (b, k)$ with $h, k > 0$.

2 The Torricelli (Fermat) point of a triangle

Lab 2 — 90 minutes. Work in pairs, then compare arguments at the board. Throughout the main activities, ABC is a nondegenerate triangle whose three angles are strictly smaller than 120° . We seek to minimize

$$F(X) = |X - A| + |X - B| + |X - C|, \quad X \in \triangle ABC,$$

where $\triangle ABC$ includes its interior and boundary.

Exercise 2.1 (Warm-up — 10 minutes). In an equilateral triangle of side s , compute the sum of distances from the center to the three vertices. Compare it with the sum at a vertex. Is comparison at these four points a proof of global minimality?

Exercise 2.2 (Construct the candidate — 20 minutes). Construct equilateral triangles externally on AB , BC and CA . Join each new vertex to the opposite vertex of ABC .

Construction fact supplied for this lab: under the angle assumption above these three segments meet at an interior point T , and $\angle ATB = \angle BTC = \angle CTA = 120^\circ$. The associated circumcircles give another way to locate T .

- (a) Make a careful drawing and measure the three angles at T .
- (b) Using an outward equilateral triangle ABD , check that a point on its circumcircle on the opposite side of AB from D sees AB at an angle of 120° . Use this to explain the angle condition in the circle construction.

We will prove global minimality of this candidate using elementary geometry. Measurements alone establish neither exact angles nor optimality.

Exercise 2.3 (An auxiliary equilateral triangle — 20 minutes). Draw lines d_A, d_B, d_C through A, B, C perpendicular to TA, TB, TC , respectively. Take the half-plane bounded by each line which contains T .

- (a) Explain why their intersection is an equilateral triangle K . Check that ABC lies inside K . You may use the fact that the angles between the three rays are 120° .
- (b) Prove Viviani's identity: the sum of the perpendicular distances from an interior point of an equilateral triangle to its three side lines is its altitude. Hint: add the areas of the three triangles with this point as their common vertex.

Exercise 2.4 (Prove the minimum — 25 minutes). Let $X \in \triangle ABC$.

- (a) Compare $|X - A|$ with $\text{dist}(X, d_A)$, and similarly for B, C .
- (b) Apply Viviani's identity at X and T to prove

$$|X - A| + |X - B| + |X - C| \geq |T - A| + |T - B| + |T - C|.$$

- (c) Show that equality forces $X = T$.

Discussion and exit task — 15 minutes. Identify precisely where the construction fact was used. State why the proof establishes a unique global minimum on the entire triangle. Sketch what happens when one angle approaches 120° . The result for an angle at least 120° is that this vertex is the minimizer; its proof is an optional extension in the instructor notes.

Optional extension. For $A = (0, 0)$, $B = (2, 0)$ and $C = (0, 2)$, verify that $T = (1 - 1/\sqrt{3}, 1 - 1/\sqrt{3})$ has the required angles and compute $F(T)$.

3 Existence, convexity and elementary minima

Lab 3 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 3.1 (Diagnose existence — 15 minutes). For each problem, find the infimum and decide whether it is attained. If it is not, give a minimizing sequence. Identify which hypothesis of the compactness/continuity argument is missing.

- (a) $f(x) = x$ on $(0, 1]$.
- (b) $f(x) = e^x$ on $\mathbb{R}$.
- (c) $f(x) = (x - 2)^2$ on $[0, 1]$.
- (d) $f(0) = 1$, $f(x) = x$ for $x \in (0, 1]$, on $[0, 1]$.

Exercise 3.2 (A rectangle of prescribed area — 25 minutes). Fix $A_0 > 0$. A rectangle has sides $a, b > 0$, area $ab = A_0$ and perimeter $2(a + b)$.

- (a) Eliminate b and write the objective as $p(a)$ on $(0, \infty)$.
- (b) Check its behavior as $a \downarrow 0$ and $a \rightarrow \infty$. Explain why one can restrict the search to a compact interval.
- (c) Compute p' and its sign; find the unique minimizer and minimum.
- (d) Prove the same global bound using $(a - b)^2 \geq 0$.

Exercise 3.3 (Nearest point to a line, using calculus — 20 minutes). Let $Q = (1, 2)$ and $\ell = \{(t, 2t) : t \in \mathbb{R}\}$. Minimize squared distance to Q along ℓ .

- (a) Explain why squaring distance does not change the minimizing point.
- (b) Compute the resulting quadratic, its derivative and its minimizer.
- (c) Change Q to $(2, 1)$ and repeat. Verify that the displacement from Q to the closest point is perpendicular to $(1, 2)$.

Exercise 3.4 (Stationary points and uniqueness — 20 minutes). (a) Compute the gradient of $f(x, y) = x^2 + 2y^2 - 2x + 4y$ and find its stationary point. Complete squares to prove global minimality and uniqueness.

- (b) Find and classify the stationary point of $q(x, y) = x^2 - y^2$ by examining the coordinate axes.
- (c) Explain why strict convexity cannot be used to prove uniqueness of the minimizer of x^2 on $\{-1, 1\}$.

Synthesis — 10 minutes. Present the rectangle solution in three stages: existence, identification, uniqueness. Which part of the argument goes beyond solving $p'(a) = 0$?

Optional extension. For two nonparallel lines $A + tu$ and $B + sv$ in $\mathbb{R}^3$, minimize $\|A + tu - B - sv\|^2$. Derive the two equations saying the joining segment is orthogonal to both directions. Explain why the quadratic in (t, s) is strictly convex when u, v are linearly independent.

4 Multipliers and polygonal isoperimetry

Lab 4 — 90 minutes. Work in pairs, then compare arguments at the board. The core goal is to see how a shape problem becomes a finite-dimensional problem, and how the multiplier equations force a regular polygon.

Exercise 4.1 (One-constraint warm-up — 15 minutes). Minimize $2(a+b)$ subject to $ab = A_0 > 0$, with $a, b > 0$. Check the constraint gradient, form the Lagrangian and solve its equations. Use Lab 3 to justify global minimality.

Exercise 4.2 (The regular candidate — 15 minutes). For a regular n -gon ($n \geq 3$) of perimeter L , let $s = L/n$ be its side length and r its apothem (distance from center to a side).

- (a) Use one of its n congruent central triangles to show $r = s/(2 \tan(\pi/n))$.
- (b) Deduce

$$A_{\text{reg}} = \frac{L^2}{4n \tan(\pi/n)}.$$

- (c) Check $n = 3, 4$ and the limit $n \rightarrow \infty$.

Setup and supplied lemma — 10 minutes

For a counterclockwise convex polygon with vertices $v_i = (x_i, y_i)$, indices taken modulo n , define

$$P = \sum_{i=1}^n \|v_{i+1} - v_i\|, \quad A = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - y_i x_{i+1}).$$

Geometric lemma supplied today. Among convex polygons with at most n vertices and perimeter $L > 0$, an area maximizer exists. It has exactly n distinct vertices, no consecutive collinear edges, and each interior angle is less than π . The instructor notes prove this lemma; the main lab focuses on the smooth optimality argument it makes possible.

Let $R(x, y) = (-y, x)$ be rotation by 90° counterclockwise. At a strictly convex polygon the gradient formulas are

$$\nabla_{v_i} P = e_{i-1} - e_i, \quad \nabla_{v_i} A = -\frac{1}{2} R(v_{i+1} - v_{i-1}), \quad e_i = \frac{v_{i+1} - v_i}{\|v_{i+1} - v_i\|}.$$

Here e_i is the oriented unit vector along edge i .

Exercise 4.3 (The multiplier equations — 20 minutes). (a) Verify the area gradient by differentiating the two shoelace terms containing v_i . Verify the perimeter gradient from the two adjacent lengths.

- (b) Under dilation $v_i \mapsto t v_i$, perimeter scales by t and area by t^2 . Differentiate at $t = 1$ to obtain

$$\sum_i \nabla_{v_i} P \cdot v_i = P, \quad \sum_i \nabla_{v_i} A \cdot v_i = 2A.$$

Why is the full constraint gradient nonzero?

- (c) Maximize area at fixed perimeter by writing $\nabla A = \lambda \nabla P$. Show that $\lambda = 2A/P > 0$.

Exercise 4.4 (Equal sides and equal angles — 20 minutes). Write $\ell_i = \|v_{i+1} - v_i\|$. Let $\theta_i \in (0, \pi)$ be the exterior turning angle between e_{i-1} and e_i .

- (a) Dot the equation at vertex i with $v_{i+1} - v_{i-1} = \ell_{i-1}e_{i-1} + \ell_ie_i$. The area-gradient contribution vanishes. Deduce

$$(\ell_{i-1} - \ell_i)(1 - \cos \theta_i) = 0,$$

so all side lengths equal some s .

- (b) Use $\|e_{i-1} - e_i\| = 2 \sin(\theta_i/2)$ and $\|e_{i-1} + e_i\| = 2 \cos(\theta_i/2)$ in the same multiplier equation to obtain $\lambda = (s/2) \cot(\theta_i/2)$. Why must all exterior angles equal $2\pi/n$?

Synthesis — 10 minutes. Combine the supplied existence lemma, the regularity of the constraint and the multiplier calculation to conclude that, for convex n -gons,

$$P^2 \geq 4n \tan(\pi/n) A,$$

with equality precisely for regular n -gons, up to rigid motions. Explain why solving the multiplier equations alone would leave a gap.

Optional extension A: two constraints. Minimize $x^2 + y^2 + z^2$ under $x + y + z = 1$ and $x - y = 1$. Check independence of the gradients, solve the Lagrange equations and prove global minimality.

Optional extension B: Fermat revisited. For $X \notin \{A, B, C\}$, compute the gradient of $F(X) = |X - A| + |X - B| + |X - C|$. Show that its vanishing gives three unit vectors summing to zero and hence three angles of 120° . Identify what fails at a vertex.

5 Scaling, regular polygons and Dido's problem

Lab 5 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 5.1 (Two families of regular polygons — 15 minutes). Using the side length and apothem, derive

$$P_n(A_0) = 2\sqrt{A_0 n \tan(\pi/n)}, \quad A_n(L) = \frac{L^2}{4n \tan(\pi/n)}.$$

Compute the values for $n = 3, 4, 6, 12$ at $A_0 = 1$ and at $L = 1$. Sketch the equal-area shapes separately from the equal-perimeter shapes. State a conjecture about the optimal region and explain why a numerical table of regular polygons cannot prove it.

Exercise 5.2 (Prove the monotonicity — 15 minutes). Put $t = \pi/n$ and $h(t) = t - \sin t \cos t$.

- (a) Show that $h'(t) = 2 \sin^2 t$ and $h(t) > 0$ for $0 < t < \pi/2$.
- (b) Deduce that $\tan t/t$ increases with t , and hence that $n \tan(\pi/n)$ decreases with n .
- (c) Prove that the two quantities in Exercise 5.1 converge to the corresponding disk values. Which one increases, and which one decreases?

Exercise 5.3 (Transfer competitors by scaling — 15 minutes). Suppose K_* has perimeter 10, area 6, and maximizes area among all admissible shapes with perimeter 10.

- (a) A proposed competitor H has area 6 and perimeter 9. Scale it to perimeter 10 and find its new area. Explain the contradiction.
- (b) Which dilation of K_* minimizes perimeter at area 24? Give its perimeter and justify the assertion using arbitrary competitors.
- (c) Explain why the same argument may fail if every shape must stay inside a fixed square.

Exercise 5.4 (The missing step between polygons and curves — 20 minutes). Let K be any convex region of perimeter L . You may use the following two established facts:

- The inscribed polygon Q_m formed by m equally spaced boundary points has at most m vertices, $P(Q_m) \leq L$, and $A(Q_m) \rightarrow A(K)$.
- Lab 4 proves $4m \tan(\pi/m) A(Q_m) \leq P(Q_m)^2$.

- (a) Combine these facts and take a limit to prove $4\pi A(K) \leq L^2$.
- (b) Explain how convexification extends the inequality to a nonconvex Jordan region.
- (c) Exhibit an attaining region. Does your limit argument alone prove that every equality case is a disk?

Exercise 5.5 (Free shoreline — 15 minutes). A fence of length 12 encloses land against a straight shore. The shore segment costs no fence, and the two endpoints may slide along it.

- (a) Reflect the region in the shore and record the new area and perimeter.
- (b) Find the maximum land area, the optimal radius and the length of the shore segment used by this design.
- (c) Compare with a closed fence of length 12 enclosing land away from the shore. Explain the factor between the two optimal areas.

Synthesis — 10 minutes. Draw the proof dependencies: polygonal existence and inequality; approximation of an arbitrary convex region; convex hull; disk attainment. Identify the separate equality-case question left for Session 6.

Optional extension. A convex boundary of length L is sampled in m arcs of length $\delta = L/m$. Explain why the cap between one arc and its chord lies in a disk of radius δ centered at the first endpoint. Deduce $0 \leq A(K) - A(Q_m) \leq \pi L^2/m$.

6 Bisecting lines, hinges and the existence gap

Lab 6 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 6.1 (What does reflection prove? — 15 minutes). Assume a convex region maximizes area A at perimeter L .

- A line bisects its area into two pieces with free arc lengths ℓ and $L - \ell$. Show that $\ell = L/2$. Include the rescaling step if the reflected piece has perimeter less than L .
- Conversely, show that a chord joining perimeter-halving points must also halve area.
- A student concludes that the two original pieces are mirror images. Explain why the preceding comparisons do not establish this conclusion.

Exercise 6.2 (Two rigid caps and a hinge — 20 minutes). A convex half-region has base endpoints A, B and a point C on its curved boundary. Its area is the area of triangle ACB plus two caps. The caps are moved rigidly while $a = |AC|$, $b = |BC|$ stay fixed.

- Express the variable part of the area in terms of $\theta = \angle ACB$ and differentiate it.
- For $a = 3$, $b = 4$ and $\theta = 60^\circ$, compute the triangle area, its right-angle value, and the possible increase in the reflected full-region area. Treat the full right-angle motion as an illustrative calculation; a small admissible motion suffices for the proof.
- Explain why maximality forces $\theta = 90^\circ$ for every C , then use Thales' theorem to identify the boundary arc.
- State what must be checked about the moved caps and the base line. Is convexity of the new competitor required in our admissible class?

Exercise 6.3 (Brahmagupta and the general formula — 20 minutes). For a convex quadrilateral $ABCD$, set $a = |AB|$, $b = |BC|$, $c = |CD|$, $d = |DA|$, $s = (a + b + c + d)/2$, $\alpha = \angle ABC$ and $\gamma = \angle CDA$. Use

$$Q^2 = (s - a)(s - b)(s - c)(s - d) - abcd \cos^2\left(\frac{\alpha + \gamma}{2}\right).$$

- Derive the area bound for fixed side lengths and characterize equality. Explain why the angle term cannot be omitted for arbitrary quadrilaterals.
- Compare a rhombus of side 2 with a 60° angle and a square of side 2. Verify the formula directly.
- A student proposes to derive the fixed-side quadrilateral bound from the planar isoperimetric inequality, then use it to prove that very inequality. Identify the circular reasoning.

Exercise 6.4 (Four hinges need only a small motion — 15 minutes). Let $t = |AC|$ be a diagonal of the quadrilateral in Exercise 6.3. The two triangle area terms are $\frac{1}{2}ab \sin \alpha$ and $\frac{1}{2}cd \sin \gamma$.

- (a) Differentiate $t^2 = a^2 + b^2 - 2ab \cos \alpha = c^2 + d^2 - 2cd \cos \gamma$ to show $dQ/dt = \frac{t}{2}(\cot \alpha + \cot \gamma)$.
- (b) Show that this derivative is nonzero for a noncyclic convex quadrilateral. Explain how this supplies a small area-increasing flex.
- (c) Explain why a strictly convex boundary that is not a circle has four boundary points forming a noncyclic quadrilateral. What stays unchanged when the four attached caps move rigidly?

Exercise 6.5 (A logical counterexample — 10 minutes). Let the candidates be the positive integers and let $J(n) = 1 - 1/n$. Consider the rule that leaves 1 unchanged and sends $n \geq 2$ to $n + 1$.

- (a) Show that every candidate except 1 is improved by this rule.
- (b) Compute $\sup J$ and decide whether it is attained. Is the distinguished candidate 1 optimal?
- (c) Explain the lesson for a geometric improvement procedure whose only unchanged shape is a disk.

Synthesis — 10 minutes. Assemble a proof of the planar inequality with its equality case. Separate the polygonal/limit argument that establishes the bound and attainment from the geometric argument that characterizes all optimizers.

Optional extension. Derive the general quadrilateral area formula from $2Q = ab \sin \alpha + cd \sin \gamma$ and the two cosine-law expressions for the diagonal. The instructor notes give a factorization that avoids a long expansion.

7 Distances between shapes and compactness

Lab 7 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 7.1 (Two metrics — 15 minutes). (a) Explain why $d_1(E, F) = |E \triangle F|$ is only a pseudometric on measurable sets. Give two distinct compact sets at distance zero. Prove $||E| - |F|| \leq d_1(E, F)$.

(b) For nonempty compact intervals, prove

$$d_H([a, b], [c, d]) = \max\{|a - c|, |b - d|\}.$$

Compute both distances between $[0, 2]$ and $[1, 3]$, using length as the one-dimensional measure in d_1 .

(c) Explain why a small directed excess $\rho(K, L)$ alone does not imply a small Hausdorff distance.

Exercise 7.2 (Independent notions of convergence — 20 minutes). Let $S = [0, 1]^2$, $K_n = S \cup ([1, 2] \times [0, 1/n])$, and $G_n = \{(i/n, j/n) : 0 \leq i, j \leq n\}$.

(a) Calculate $d_1(K_n, S)$ and $d_H(K_n, S)$.

(b) Calculate $d_1(G_n, S)$ and $d_H(G_n, S)$. For the latter, locate the points farthest from the grid.

(c) Which implication between the two convergences does each example disprove? Explain the consequence for area continuity.

(d) Find the actual Hausdorff limit of K_n . Does it have the same area as S ?

Exercise 7.3 (An intersection can jump — 10 minutes). Let $L = [(0, 0), (1, 0)]$ and $K_n = [(0, 0), (1, 1/n)]$, where $[A, B]$ denotes a segment.

(a) Show that $K_n \rightarrow L$ in Hausdorff distance.

(b) Compute $K_n \cap L$ and compare its limit with $L \cap L$.

(c) Both sequences are convex and every intersection is nonempty. Explain why these facts do not make intersection continuous.

Exercise 7.4 (Distance functions and geometric properties — 20 minutes). For nonempty compact sets $K, L \subset D$, with D compact:

- (a) Prove that d_K is 1-Lipschitz.
- (b) Prove $|d_K(x) - d_L(x)| \leq d_H(K, L)$, then evaluate at points of K and L to obtain $\|d_K - d_L\|_{L^\infty(D)} = d_H(K, L)$. Here the norm means the supremum on D , not an essential supremum.
- (c) Approximate endpoints of a diameter to prove $|\text{diam } K - \text{diam } L| \leq 2d_H(K, L)$.
- (d) Prove that a Hausdorff limit of convex compact sets is convex. Identify exactly where closedness of the limit is used.

Exercise 7.5 (Complete the selection proof — 15 minutes). Suppose Arzelà–Ascoli has supplied uniform convergence $d_{K_n}|_D \rightarrow f$, and set $K = \{f = 0\} \subset D$.

- (a) Choose $x_n \in K_n$ to show that K is nonempty.
- (b) Use the Lipschitz bound to prove $f \leq d_K$.
- (c) For fixed $x \in D$, choose nearest points $y_n \in K_n$ and a convergent subsequence. Prove $f(x) \geq d_K(x)$.
- (d) Explain why the common uniformly convergent subsequence converges to K in d_H , even though the nearest-point subsequence in (c) may depend on x .

Synthesis — 10 minutes. Diagnose these two proposed existence arguments: “All competitors are convex bodies, so the limit has interior”; “All competitors have area one, so Blaschke applies without any translation or diameter estimate.” Give a counterexample to each inference and name what an actual existence proof must check after extracting a subsequence.

Optional extensions. Prove that a decreasing sequence of nonempty compact subsets of D converges to its intersection. Prove that a Hausdorff limit of connected compact sets is connected. Does the grid example contradict upper semicontinuity of area, namely $\limsup_n |K_n| \leq |K|$? Prove this inequality for compact sets in a fixed bounded container using $K_n \subset K_\varepsilon$.

8 Support functions, existence and minimum width

Lab 8 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 8.1 (Support and width — 15 minutes). (a) Find the support function of the disk centered at $(2, 0)$ with radius 1. Find a direction where it is negative. Compute its directional widths.

- (b) For $K = [-a, a] \times [-b, b]$, derive $h_K(\theta)$ and show $w(K) = 2 \min(a, b)$ and $\text{diam } K = 2\sqrt{a^2 + b^2}$.
- (c) Explain why a circle and its filled disk have the same support function but positive Hausdorff distance. Which hypothesis is missing?
- (d) Compare a disk of diameter 1, a Reuleaux triangle of width 1, and an equilateral triangle of altitude 1. You may use the stated constant-width property of the Reuleaux triangle. Which shapes have constant width, and which have minimum width 1?

Exercise 8.2 (Continuity and admissible limits — 15 minutes). Assume K_n, K are planar compact convex sets and $d_H(K_n, K) = \varepsilon_n \rightarrow 0$.

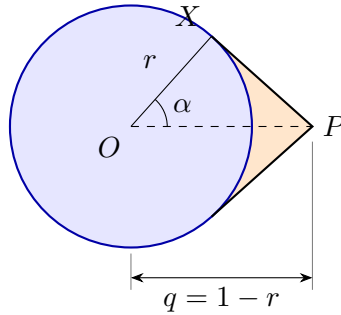
- (a) Prove $|w(K_n) - w(K)| \leq 2\varepsilon_n$ using support functions. Explain why both $w(K) = 1$ and constant width 1 define closed classes.
- (b) If $\overline{B}(0, r) \subset K$, use the two homothetic inclusions from the course to bound $|K_n \triangle K|$ by $4\varepsilon_n A(K)/r$ for $\varepsilon_n < r$.
- (c) Compute the area and perimeter of $[0, 1] \times [0, 1/n]$. What value must the continuous extension of perimeter assign to the limiting unit segment? Explain why width at least 1 excludes this degeneration in the plane.

Exercise 8.3 (Find the missing confinement — 15 minutes). We minimize area among planar compact convex sets with $w(K) \geq 1$.

- (a) Give an admissible sequence with unbounded diameter.
- (b) For a diameter segment AB , draw the two triangles joining AB to support points on opposite sides. Deduce $A(K) \geq \text{diam}(K)w(K)/2$.
- (c) Use an equilateral triangle to restrict a minimizing sequence to $A(K_n) \leq 2/\sqrt{3}$. Translate, extract a subsequence, pass the constraint and objective to the limit, and conclude existence.
- (d) Explain why every minimizer has $w(K) = 1$.

Exercise 8.4 (An incircle and three tangent regions — 20 minutes). Normalize $w(K) = 1$, and take a largest inscribed disk of radius r . *Supplied geometric facts:* $1/3 \leq r \leq 1/2$. If $r = 1/3$, K is equilateral. If $r = 1/2$, the disk alone gives $A(K) \geq \pi/4$. For $1/3 < r < 1/2$, there are three points in K at distance $q = 1 - r$ from the disk center such that their tangent regions outside the disk have disjoint interiors. The instructor notes justify these facts without assuming equal angular spacing.

Study one of the three congruent tangent regions. Let O be the disk center, P the external point, X a tangency point, and $\alpha = \angle XOP$.



- (a) Use $OX \perp PX$ to compute PX and $\cos \alpha$.
- (b) Subtract a circular sector from two right triangles to show that the added area is $r\sqrt{1-2r} - r^2\alpha$.
- (c) Add the disk and three tangent regions to obtain

$$A(K) \geq F(r) = \pi r^2 + 3r\sqrt{1-2r} - 3r^2 \arccos\left(\frac{r}{1-r}\right).$$

Why is disjointness outside the disk essential to this estimate?

Exercise 8.5 (Finish Pál's inequality — 15 minutes). Use the derivative identity

$$F'(r) = 6r \left[\frac{\pi}{3} - \arccos\left(\frac{r}{1-r}\right) \right] + \frac{3(1-2r)^{3/2}}{1-r}.$$

- (a) Prove $F'(r) > 0$ for $1/3 < r < 1/2$ and calculate $F(1/3)$.
- (b) Conclude $A(K) \geq 1/\sqrt{3}$, including $r = 1/2$, and identify equality using the supplied geometric facts.
- (c) Rescale to minimum width w_0 . Give the optimal area and the side length of the optimizer.

Synthesis — 10 minutes. Present the existence proof and the identification proof separately. Why does the triangle result not solve area minimization among constant-width bodies?

Optional extensions. Verify the derivative formula. Prove Cauchy's perimeter formula for a convex polygon by integrating projected side lengths; deduce that every planar constant-width- w_0 body has perimeter πw_0 . Use the contact triangle in the instructor notes to prove $r \geq w(K)/3$ and analyze equality.

9 Minkowski sums, mixed areas and Brunn–Minkowski

Lab 9 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 9.1 (Compute a sum, identify its cross term — 15 minutes). Let $K = [0, 2] \times [0, 1]$ and $L = [0, 1] \times [0, 3]$.

- (a) Draw $K, L, K + L$. Compute $A(K + tL)$ and identify $V(K, L)$. Explain why this number is not $A(K \cap L)$.
- (b) Prove $h_{K+L} = h_K + h_L$ directly from the definition. Deduce $V(K, K) = A(K)$ using $K + K = 2K$.
- (c) Show that $h_{K+(-K)}(u) = w_K(u)$. If a convex body has constant width w , identify its difference body and state its radius.

Exercise 9.2 (A parallel boundary — 15 minutes). Let B be the unit disk and use the rectangle K above.

- (a) Draw $K + tB$, dividing the added area into four strips and four quarter-disks. Compute its area and perimeter.
- (b) Calculate both quantities at $t = 1/2$, and compute $V(K, B)$. Explain geometrically why the area derivative at $t = 0$ is $P(K)$.
- (c) Two unit disks have centers distance 10 apart. For $0 < t < 1$, compute the area of the parallel set of their union. Why is the single πt^2 formula not valid in this example?

Exercise 9.3 (The matching-proportion cut — 20 minutes). Let

$$E = ([0, 1] \times [0, 1]) \cup ([2, 3] \times [0, 1]), \quad F = [0, 2] \times [0, 1].$$

- (a) Cut E at $x = 3/2$. Find the parallel cut in F giving the same area fraction on its left. Write the four pieces explicitly.
- (b) Compute $E^- + F^-$ and $E^+ + F^+$. Where is their separating line? Add their areas and compare with the Brunn–Minkowski lower bound.
- (c) Compute the full sum $E + F$. Is it exhausted by the two sums in (b)? Why does the proof need only an inclusion?
- (d) In the general induction, the cut can split additional boxes. Explain why each pair still has a strictly smaller total box count, even if the number of pieces across *both* sides increases.

Exercise 9.4 (Limits and inequalities — 20 minutes). (a) Let E_j, F_j be outer grid-box approximations with $E \subset E_j \subset E + \delta_j B$ and $F \subset F_j \subset F + \delta_j B$, where $\delta_j \rightarrow 0$. Bound $E_j + F_j$ between two sets and justify volume convergence. Why would citing general Hausdorff continuity of volume be incorrect?

(b) Compare $A(K+tL)$ with $(\sqrt{A(K)} + t\sqrt{A(L)})^2$ to prove $V(K, L)^2 \geq A(K)A(L)$. Check the inequality for the rectangles in Exercise 9.1. State the equality condition for positive-area convex bodies, using the rigidity theorem supplied in the course.

(c) Set $L = B$ and deduce $P(K)^2 \geq 4\pi A(K)$. For a general fixed convex body L , what shape minimizes $P_L(K) = 2V(K, L)$ at prescribed area A_0 ?

Exercise 9.5 (Three-dimensional coefficients — 10 minutes). Let $K = [0, 1]^3$ and $L = [0, 2] \times [0, 3] \times [0, 4]$.

(a) Expand $|K + tL|$. Identify $V(K, K, L)$ and $V(K, L, L)$, including the binomial factors.

(b) Check $V(K, K, L)^3 \geq |K|^2|L|$. Is equality expected for these boxes?

(c) Using $\kappa_3 = 4\pi/3$ and $S(K) = 3V(K, K, B)$, convert the general isoperimetric inequality into $S(K)^3 \geq 36\pi|K|^2$.

Synthesis — 10 minutes. Give the logical chain from the box inequality to planar isoperimetry. Identify which results are proved, which are passed to the limit, and which equality statements are supplied.

Optional extension: the isodiametric inequality. For a compact convex body of diameter D , prove $K + (-K) \subset DB$. Use Brunn–Minkowski to show $|K| \leq \kappa_d(D/2)^d$. In the plane, apply this to a body of constant width w to identify the maximal area and an attaining shape.

Python supplement. Two notebook exercises compute sums of convex polygons and visualize the box cut with exact rectangle-union areas.

10 Constant width, support functions and a sharp mixed-area bound

Lab 10 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 10.1 (Three descriptions of the same constraint — 15 minutes). Let K be a compact convex body and B the unit disk.

- (a) Prove that constant width w , $h_K(u) + h_K(-u) = w$, and $K + (-K) = wB$ are equivalent. Explain the role of convexity.
- (b) Identify the disk of width one and its difference body. Show that a centrally symmetric constant-width body must be a disk.
- (c) Prove that $(1 - t)K + tL$ has width one if K, L do and $0 \leq t \leq 1$. Does its area have to be the linear interpolation of their areas?

Exercise 10.2 (A smooth family, with a convexity constraint — 20 minutes). Consider $h_a(\theta) = 1/2 + a \cos 3\theta$.

- (a) Verify the width identity. Compute $h_a + h_a''$ and find the exact feasible interval for a .
- (b) Use $x = h_a u + h_a' u^\perp$ to write the boundary coordinates. Find its area and perimeter using the formulas in the course.
- (c) Minimize area in this one-parameter family. Compare with the disk and the Reuleaux triangle. Why is the family minimum not the solution of the unrestricted constant-width problem?
- (d) What does adding $b \cos \theta + c \sin \theta$ do to the body, its width, its curvature radius and its area?

Exercise 10.3 (Existence and a hexagon — 20 minutes). (a) Prove existence of an area minimizer at constant width one: state confinement, compactness, closure and objective continuity separately.

- (b) For $m(\theta) = h(\theta) - 1/2$, prove that $F(\theta) = m(\theta) - m(\theta + \pi/3) + m(\theta + 2\pi/3)$ changes sign under a shift by $\pi/3$.
- (c) At a zero of F , solve the first two strip-midline equations for their common center c , and explain why the third follows. Compute the side, perimeter and area of the resulting hexagon.

Exercise 10.4 (Mixed area can stay unchanged under enlargement — 25 minutes). Let T be an equilateral triangle of side one, centered at zero. Let R be its Reuleaux triangle and $H = \text{conv}(R \cup (-R))$.

- (a) Draw these three shapes. Compute $A(T)$, $A(R)$ and $A(H)$. Check that H is a regular hexagon with alternating vertices from T and $-T$.
- (b) Draw $T + (-T)$ and compute $V(T, -T)$. Use monotonicity to show that $V(L, -L)$ has the same value for every convex $T \subset L \subset H$.
- (c) Starting from $A(K + (-K)) = \pi$ for a width-one body, give Chakerian's proof. Identify the separate geometric input.
- (d) Explain why equality in mixed-area monotonicity does not imply equality of the two bodies. Obtain the maximal area of a width-one body using Barbier and isoperimetry.

Synthesis — 10 minutes. Compare the minimum-width problem from Session 8 with today's constant-width problem: admissible class, confinement, minimizer and minimum value.

Python supplement. Reconstruct bodies from odd Fourier modes and check convexity; numerically locate a circumscribed regular hexagon and compare mixed areas of a triangle, Reuleaux triangle and hexagon.

11 Tangent polygons and width-preserving variations

Lab 11 — 90 minutes. Work in pairs, then compare arguments at the board.

Exercise 11.1 (Two corner cuts — 20 minutes). In a tangent polygon with normal spacing δ , use the opposite rhombus construction from the lecture. The new cuts have depths $t_1, t_2 \geq 0$ with $t_1 + t_2 = s = w(\sec(\delta/2) - 1)$.

- (a) Draw the two removed triangles. Derive their combined area $D = \cot(\delta/2)(t_1^2 + t_2^2)$.
- (b) Find the extreme values of D at fixed s , including equality conditions. Explain the disk and Reuleaux-triangle configurations.
- (c) Starting with $\delta = \pi/3$ and $w = 1$, compute s and the total area loss when passing from 6 to 12 tangent directions around the disk and around the Reuleaux triangle.

Exercise 11.2 (A strict gap that survives the limit — 20 minutes). Let $Q_N(K)$ be the tangent polygons with $N = 6 \cdot 2^j$ directions, starting at a regular circumscribed hexagon.

- (a) Use the cut bounds to prove the area comparison between the Reuleaux triangle, K , and the disk by induction.
- (b) Explain why a positive gap at one stage cannot disappear in the limit. What is missing from the bare assertion “all finite approximations have strictly greater area”?
- (c) Prove that an interior corner angle of a constant-width body is at least 120° . Explain why a 120° corner forces a Reuleaux triangle, and finish the lower equality case.

Exercise 11.3 (Regular Reuleaux polygons — 20 minutes). Let $n \geq 3$ be odd. The centers of the regular width-one Reuleaux n -gon form a regular polygon of circumradius $r_n = 1/(2 \cos(\pi/(2n)))$.

- (a) Check that its longest diagonals have length one. Its n boundary arcs have radius one and angle π/n . Add the skeleton area and the n circular-segment areas to obtain $A_n = \pi/2 - (n/2) \tan(\pi/(2n))$.
- (b) Differentiate the real-variable extension in n , or study $\tan t/t$, to prove strict increase. Compute A_3 and the limit.
- (c) Why does comparing regular examples not prove the full Blaschke–Lebesgue theorem? Supply the missing stationarity, existence and approximation steps.

Exercise 11.4 (A constrained gradient — 20 minutes). For a nondegenerate Reuleaux n -gon, $n = 2k + 1 \geq 5$, use the dependent-circle construction from the lecture and the gradient in the course notes. For use without the course PDF, the formula is

$$g_0 = (\tan(\theta_k/2) - \tan(\theta_0/2))(x_0 - x_k) + (\tan(\theta_{k+1}/2) - \tan(\theta_0/2))(x_0 - x_{k+1}).$$

- (a) If $g_0 \neq 0$, propose an area-decreasing initial velocity. Explain why x_k, x_{k+1} must also move, and why the step must be small.
- (b) If $g_0 = 0$, use independence of the two diameter vectors to identify the three arc angles. Repeat around the polygon.
- (c) For $n = 5$, a regular example has $g_0 = 0$. Does this make it a minimum? Compare A_5 with A_3 . As a stronger local test, evaluate the sign of the supplied second variation

$$A''(0) = 2 \frac{\sin(\theta/2) \sin \theta - \cos(3\theta/2)}{\cos(\theta/2) \sin \theta}, \quad \theta = \pi/5.$$

- (d) Why must the compact existence class allow *at most* N vertices, instead of exactly N distinct vertices?

Synthesis — 10 minutes. Draw the logical dependencies of the mixed-area, tangent-polygon and variational proofs. State which argument supplies uniqueness and which auxiliary facts the variational proof needs.

Optional multiplier task. Set $C_i = (|x_i - x_{i+k}|^2 - 1)/2$. Compute the nonzero blocks of ∇C_i and write the stationarity equation for a Lagrangian $D + \sum_i \lambda_i C_i$, where D is the disk-polygon area.

Python supplement. Refine equiangular tangent polygons and measure the two cut extremes; perturb a Reuleaux pentagon, check its area gradient by finite differences, and observe an admissible area decrease.